

15.Design a C program to organize the file using a two-level directory structure.

AIM

To write a C program to implement a **two-level directory structure** with separate directories for different users.

CODE

```
#include <stdio.h>

}

else if(choice == 2)

{

    printf("Enter username: ");

    scanf("%s", username);

    u = -1;

    for(i = 0; i < userCount; i++)

    {

        if(strcmp(users[i], username) == 0)

        {

            u = i;

            break;

        }

    }

    if(u == -1)

    {

        printf("User not found.\n");

    }

    else

    {
```

```

    printf("Enter file name: ");
    scanf("%s", filename);

    strcpy(files[u][fileCount[u]], filename);
    fileCount[u]++;

    printf("File created successfully.\n");
}
}
else if(choice == 3)
{
    for(i = 0; i < userCount; i++)
    {
        printf("\nUser: %s\n", users[i]);

        for(int j = 0; j < fileCount[i]; j++)
            printf("%s\n", files[i][j]);
    }
}
else if(choice == 4)
{
    printf("Enter username: ");
    scanf("%s", username);

    u = -1;

    for(i = 0; i < userCount; i++)
    {
        if(strcmp(users[i], username) == 0)

```

```

    {
        u = i;
        break;
    }
}

if(u == -1)
{
    printf("User not found.\n");
}
else
{
    printf("Enter file name: ");
    scanf("%s", filename);

    int found = 0;

    for(i = 0; i < fileCount[u]; i++)
    {
        if(strcmp(files[u][i], filename) == 0)
        {
            found = 1;
            break;
        }
    }

    if(found)
        printf("File found.\n");
    else

```

```

        printf("File not found.\n");
    }
}
else
{
    break;
}
}
return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 #include <string.h> 3 4 char users[10][20]; 5 char files[10][20][30]; 6 int fileCount[10]; 7 8 int main() 9 { 10 int userCount = 0; 11 int choice, u, i; 12 char username[20], filename[30]; 13 14 while(1) 15 { 16 printf("\n--- TWO LEVEL DIRECTORY ---\n"); 17 printf("1. Create User\n"); 18 printf("2. Create File\n"); 19 printf("3. Display Files\n"); 20 printf("4. Search File\n"); 21 printf("5. Exit\n"); 22 printf("Enter choice: "); 23 scanf("%d", &choice); 24 25 if(choice == 1) 26 { 27 printf("Enter username: "); 28 scanf("%s", username); 29 30 strcpy(users[userCount], username); 31 fileCount[userCount] = 0; 32 userCount++; 33 34 printf("User created successfully.\n"); 35 } 36 else if(choice == 2) </pre>		<pre> --- TWO LEVEL DIRECTORY --- 1. Create User 2. Create File 3. Display Files 4. Search File 5. Exit Enter choice: 2 Enter username: karthi User not found. --- TWO LEVEL DIRECTORY --- 1. Create User 2. Create File 3. Display Files 4. Search File 5. Exit Enter choice: 3 --- TWO LEVEL DIRECTORY --- 1. Create User 2. Create File 3. Display Files 4. Search File 5. Exit Enter choice: 4 Enter username: kavi User not found. --- TWO LEVEL DIRECTORY --- 1. Create User 2. Create File 3. Display Files 4. Search File 5. Exit Enter choice: 5 </pre>

RESULT

Thus, the C program to organize files using a **two-level directory structure** was successfully implemented and executed.